

Hair Care Heat-Safety Station

Problem

Conventional curling irons and straighteners create significant risks of burns, hair damage, and property damage. The report cites tools reaching temperatures up to 245°C, while hair can experience significant structural damage around 200°C and protein denaturation at 237°C. It also cites an estimated 105,081 hair-care-product injuries over a five-year period, including 82,151 injuries involving curling irons. Existing alternatives are either expensive, such as the approximately \$599 Dyson Airwrap, or time-intensive, creating a need for a safer, affordable, and convenient hair-styling support system.

Process

The team researched stakeholder needs, evaluated 5 design concepts, and used a decision matrix scoring viability, feasibility, desirability, sustainability, and safety on a 1-5 scale. The selected station concept scored 21.5/25. The prototype was organized into cooling, command/control, heating/display, timer, and housing subsystems. Cooling materials were tested across 3 heat tools and 5 configurations; combining Teflon, aluminum foil, and a fan reduced cool-to-touch times to 0:48-0:58, compared with 10:01-11:15 without cooling materials.

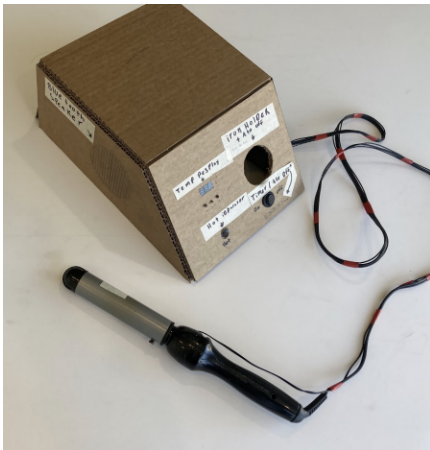


Figure 1: Alpha Prototype

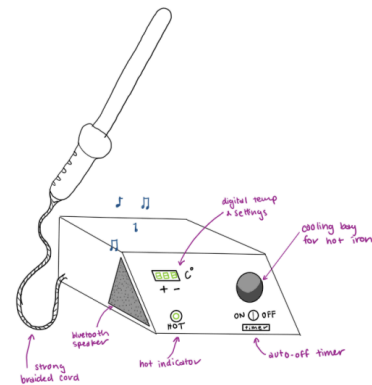


Figure 2: Alpha Prototype Concept Art

Product

The final product is a compact heat-safety station designed to house, control, and cool a curling iron or straightener. The proposed full-scale station is approximately 12 inches long and 4 pounds, with a maximum temperature of 150°C, fan and heat-indicator activation at 40°C, temperature controls, a timer/auto-off system, a non-slip base, and heat-resistant housing. Estimated full-scale materials cost \$20.41, overhead and manufacturing cost \$6.12, and target consumer price \$66.31, approximately 90% less than a \$599 air-curler alternative.

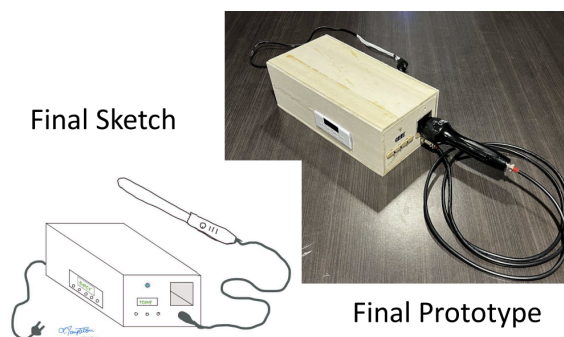


Figure 3: Final Sketch and Beta Prototype



Figure 4: Internal Electrical Components